Rayleigh Flow

Escoamento Compressível com Transferência de Calor

$$\frac{P_t}{P_t^{¿}} = \frac{\gamma+1}{1+\gamma M^2} \left\{ \left(\frac{2}{\gamma+1} \right) \left[1 + \frac{(\gamma-1)}{2} M^2 \right] \right\}^{\gamma/(\gamma-1)}$$

$$\frac{T_t}{T_t^{¿}} = \frac{2(\gamma+1)M^2}{(1+\gamma M^2)^2} \left[1 + \frac{(\gamma-1)}{2} M^2 \right]$$

Função objetivo para achar Mach:

$$F = \frac{T_t}{T_t^{¿}} - \frac{2(\gamma+1)M^2}{(1+\gamma M^2)^2} \left[1 + \frac{(\gamma-1)}{2} M^2 \right] = \frac{T_t}{T_t^{¿}} - \frac{AC}{B} = 0$$

$$\frac{dF}{dM} = \frac{-1}{B^2} \left(BC \frac{dA}{dM} + BA \frac{dC}{dM} - AC \frac{dB}{dM} \right)$$

$$A = 2(\gamma+1) M^2$$

$$B = (1+\gamma M^2)^2$$

$$C = 1 + \frac{(\gamma-1)}{2} M^2$$

$$\frac{dA}{dM} = 4(\gamma+1) M$$

$$\frac{dB}{dM} = 4\gamma M (1+\gamma M^2)$$

$$\frac{dC}{dM} = (\gamma-1) M$$